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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 20th July Morning Shift

JEE Main Question Paper with Answer Key

20 Jul 2021	90	19	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

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Out of syllabus	19 questions
Answer key	at the end

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Chemistry 21 questions

Q1. MCQ (Single Correct Answer)

Open solution

According to the valence bond theory the hybridization of central metal atom is dsp^2 for which one of the following compounds?

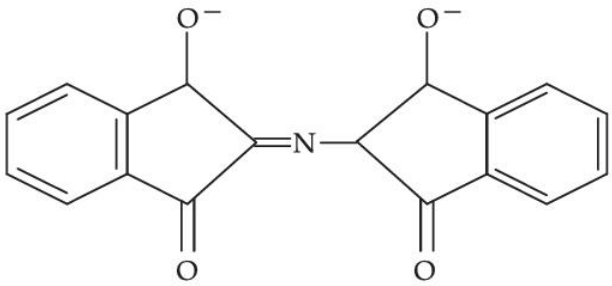
(A) $NiCl_2.6H_2O$ (B) $K_2[Ni(CN)_4]$ (C) $[Ni(CO)_4]$ (D) $Na_2[NiCl_4]$

Q2. MCQ (Single Correct Answer)

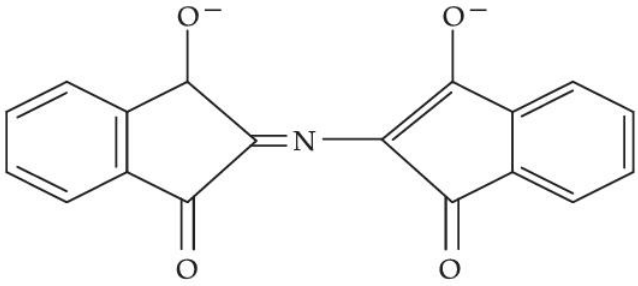
Open solution

The correct structure of Ruhemann's Purple, the compound formed in the reaction of ninhydrin with proteins is :

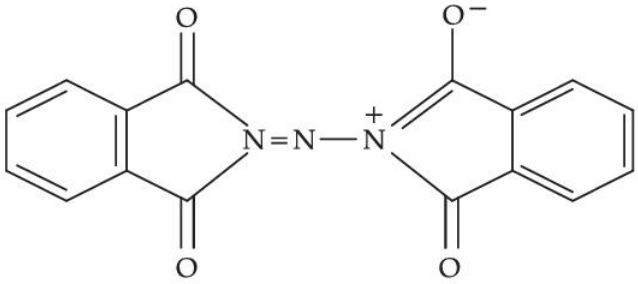
(A)



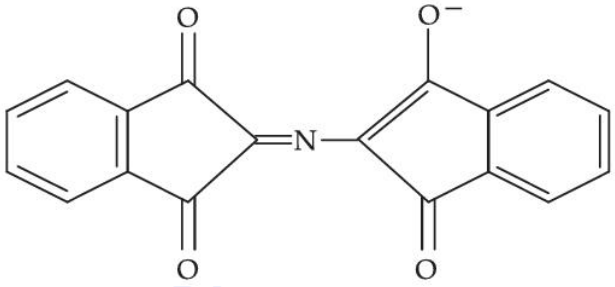
(B)



(C)



(D)



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Q3. MCQ (Single Correct Answer)

The correct order of intensity of colors of the compounds is :

(A) $[Ni(CN)_4]^{2-} > [NiCl_4]^{2-} > [Ni(H_2O)_6]^{2+}$

(B) $[Ni(H_2O)_6]^{2+} > [NiCl_4]^{2-} > [Ni(CN)_4]^{2-}$

(C) $[NiCl_4]^{2-} > [Ni(H_2O)_6]^{2+} > [Ni(CN)_4]^{2-}$

(D) $[NiCl_4]^{2-} > [Ni(CN)_4]^{2-} > [Ni(H_2O)_6]^{2+}$

Open solution

Q4. MCQ (Single Correct Answer)

The species given below that does NOT show disproportionation reaction is :

(A) BrO_4^- (B) BrO^- (C) BrO_2^- (D) BrO_3^-

Open solution

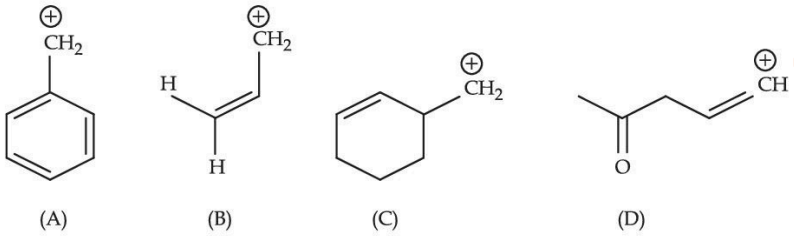
Q5. MCQ (Single Correct Answer)

An inorganic Compound 'X' on treatment with concentrated H_2SO_4 produces brown fumes and gives dark brown ring with $FeSO_4$ in presence of concentrated H_2SO_4. Also Compound 'X' gives precipitate 'Y', when its solution in dilute HCl is treated with H_2S gas. The precipitate 'Y' on treatment with concentrated HNO_3 followed by excess of NH_4OH further gives deep blue coloured solution, Compound 'X' is :

(A) $Co(NO_3)_2$ (B) $Pb(NO_2)_2$ (C) $Cu(NO_3)_2$ (D) $Pb(NO_3)_2$

Open solution

Q6. MCQ (Single Correct Answer)



(A) (B) (C) (D)

Among the given species the Resonance stabilised carbocations are :

(A) (C) and (D) only

(B) (A), (B) and (D) only

(C) (A) and (B) only

(D) (A), (B) and (C) only

Open solution

Q7. MCQ (Single Correct Answer)

In the given reaction

3-Bromo-2, 2-dimethyl butane $\xrightarrow{C_2H_5OH} \underset{(Major\ Product)}{'A'}$

Product A is :

(A) 2-Ethoxy-3, 3-dimethyl butane

(B) 1-Ethoxy-3, 3-dimethyl butane

(C) 2-Ethoxy-2, 3-dimethyl butane

(D) 2-Hydroxy-3, 3-dimethyl butane

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Q8. MCQ (Single Correct Answer)

Open solution

Compound A is converted to B on reaction with $CHCl_3$ and KOH. The compound B is toxic and can be decomposed by C. A, B and C respectively are :

(A) primary amine, nitrile compound, conc. HCl

(B) secondary amine, isonitrile compound, conc. NaOH

(C) primary amine, isonitrile compound, conc. HCl

(D) secondary amine, nitrile compound, conc. NaOH

Q9. MCQ (Single Correct Answer)

Open solution

Identify the incorrect statement from the following

(A) Amylose is a branched chain polymer of glucose

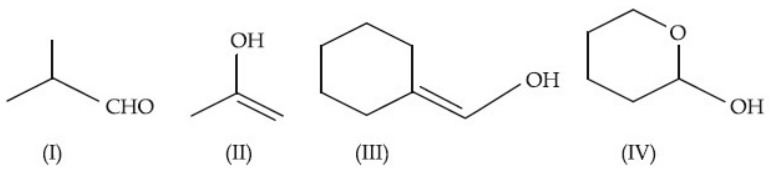
(B) Starch is a polymer of α-D glucose

(C) β-Glycosidic linkage makes cellulose polymer

(D) Glycogen is called as animal starch

Q10. MCQ (Single Correct Answer)

Open solution



(I) (II) (III) (IV)

Which among the above compound/s does/do not form Silver mirror when treated with Tollen's reagent?

(A) (I), (III) and (IV) only

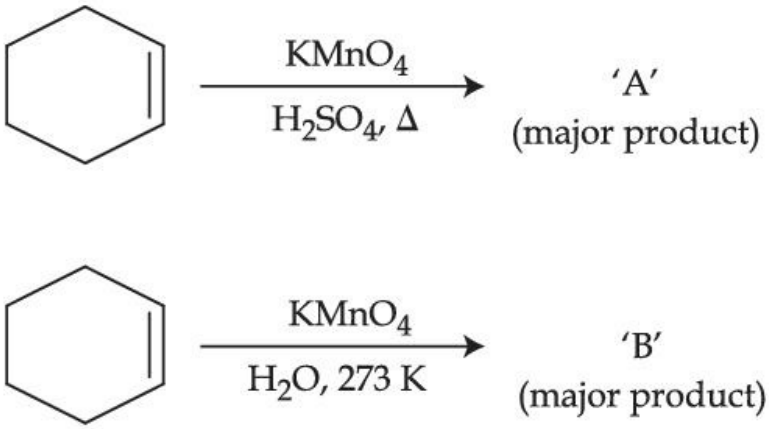
(B) Only (IV)

(C) Only (II)

(D) (III) and (IV) only

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Q11. MCQ (Single Correct Answer)



Open solution

For above chemical reactions, identify the correct statement from the following :

(A) Both compound 'A' and compound 'B' are dicarboxylic acids

(B) Both compound 'A' and compound 'B' are diols

(C) Compound 'A' is diol and compound 'B' is dicarboxylic acid

(D) Compound 'A' is dicarboxylic acid and compound 'B' is diol

Q12. Numerical

The number of lone pairs of electrons on the central I atom in I_3^- is _____.

Open solution

Q13. Numerical

250 mL of 0.5 M NaOH was added to 500 mL of 1 M HCl. The number of unreacted HCl molecules in the solution after complete reaction is _____ $\times 10^{21}$. (Nearest integer)

($N_A = 6.022 \times 10^{23}$)

Open solution

Q14. Numerical

The Azimuthal quantum number for the valence electrons of Ga^+ ion is _____.

(Atomic number of Ga = 31)

Open solution

Q15. Numerical

The spin-only magnetic moment value for the complex $[Co(CN)_6]^{4-}$ is _____ BM.

[At. no. of Co = 27]

Open solution

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Q16. Numerical

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

In an equilibrium mixture, the partial pressures are

P_{SO_3} = 43 kPa; P_{O_2} = 530 Pa and P_{SO_2} = 45 kPa. The equilibrium constant K_P = _____ $\times 10^{-2}$. (Nearest integer)

Open solution

Q17. Numerical

The number of nitrogen atoms in a semicarbazone molecule of acetone is _____.

Open solution

Q18. Numerical

To synthesize 1.0 mole of 2-methylpropan-2-ol from Ethylethanoate _____ equivalents of CH_3MgBr Reagent will be required. (Integer value)

Open solution

Q19. Numerical

The inactivation rate of a viral preparation is proportional to the amount of virus. In the first minute after preparation, 10% of the virus is inactivated. The rate constant for viral inactivation is _____ $\times 10^{-3}$ min^{-1}. (Nearest integer)

[Use : ln 10 = 2.303; $\log_{10} 3$ = 0.477; property of logarithm : $\log x^y = y \log x$]

Open solution

Q20. Numerical

An average person needs about 10000 kJ energy per day. The amount of glucose (molar mass = 180.0 g mol^{-1}) needed to meet this energy requirement is _____ g.

(Use : $\Delta_C H$(glucose) = -2700 kJ mol^{-1})

Open solution

Q21. Numerical

At 20° C, the vapour pressure of benzene is 70 torr and that of methyl benzene is 20 torr. The mole fraction of benzene in the vapour phase at 20°C above an equimolar mixture of benzene and methyl benzene is _____ $\times 10^{-2}$. (Nearest integer)

Open solution

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Mathematics 23 questions

Q22. MCQ (Single Correct Answer)

Let a be a positive real number such that $\int_0^a e^{x-[x]}dx = 10e - 9$ where [x] is the greatest integer less than or equal to x. Then a is equal to:

(A) $10 - \log_e(1+e)$ (B) $10 + \log_e 2$ (C) $10 + \log_e 3$ (D) $10 + \log_e(1+e)$

Open solution

Q23. MCQ (Single Correct Answer)

The mean of 6 distinct observations is 6.5 and their variance is 10.25. If 4 out of 6 observations are 2, 4, 5 and 7, then the remaining two observations are :

(A) 10, 11 (B) 3, 18 (C) 8, 13 (D) 1, 20

Open solution

Q24. MCQ (Single Correct Answer)

The value of the integral $\int_{-1}^{1} \log_e(\sqrt{1-x} + \sqrt{1+x})dx$ is equal to:

(A) $\frac{1}{2}\log_e 2 + \frac{\pi}{4} - \frac{3}{2}$ (B) $2\log_e 2 + \frac{\pi}{4} - 1$

(C) $\log_e 2 + \frac{\pi}{2} - 1$ (D) $2\log_e 2 + \frac{\pi}{2} - \frac{1}{2}$

Open solution

Q25. MCQ (Single Correct Answer)

If α and β are the distinct roots of the equation $x^2 + (3)^{1/4}x + 3^{1/2} = 0$, then the value of $\alpha^{96}(\alpha^{12} - 1) + \beta^{96}(\beta^{12} - 1)$ is equal to :

(A) 56×3^{25} (B) 56×3^{24} (C) 52×3^{24} (D) 28×3^{25}

Open solution

Q26. MCQ (Single Correct Answer)

Let $A = \begin{bmatrix} 2 & 3 \\ a & 0 \end{bmatrix}$, a∈R be written as P + Q where P is a symmetric matrix and Q is skew symmetric matrix. If det(Q) = 9, then the modulus of the sum of all possible values of determinant of P is equal to :

(A) 36 (B) 24 (C) 45 (D) 18

Open solution

Q27. MCQ (Single Correct Answer)

If z and ω are two complex numbers such that $|z\omega| = 1$ and $\arg(z) - \arg(\omega) = \frac{3\pi}{2}$, then $\arg\left(\frac{1-2\bar{z}\omega}{1+3\bar{z}\omega}\right)$ is :

(Here arg(z) denotes the principal argument of complex number z)

(A) $\frac{\pi}{4}$ (B) $-\frac{3\pi}{4}$ (C) $-\frac{\pi}{4}$ (D) $\frac{3\pi}{4}$

Open solution

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Q28. MCQ (Single Correct Answer)

Let [x] denote the greatest integer $\leq$ x, where x $\in$ R. If the domain of the real valued function $f(x)=\sqrt{\frac{[[x]-2}{[[x]-3}}$ is $(-\infty, a) \cup [b, c) \cup [4, \infty)$, a < b < c, then the value of a + b + c is :

(A) 8 (B) 1 (C) -2 (D) -3

Open solution

Q29. MCQ (Single Correct Answer)

Let y = y(x) be the solution of the differential equation $x \tan\left(\frac{y}{x}\right)dy = \left(y\tan\left(\frac{y}{x}\right) - x\right)dx, -1 \leq x \leq 1$, $y\left(\frac{1}{2}\right) = \frac{\pi}{6}$. Then the area of the region bounded by the curves x = 0, $x = \frac{1}{\sqrt{2}}$ and y = y(x) in the upper half plane is :

(A) $\frac{1}{8}(\pi - 1)$ (B) $\frac{1}{12}(\pi - 3)$ (C) $\frac{1}{4}(\pi - 2)$ (D) $\frac{1}{6}(\pi - 1)$

Open solution

Q30. MCQ (Single Correct Answer)

Let $A = [a_{ij}]$ be a 3 $\times$ 3 matrix, where $a_{ij} = \begin{cases} 1 & , \quad if\ i = j \\ -x & , \quad if\ |i-j| = 1 \\ 2x+1 & , \quad otherwise. \end{cases}$

Let a function f : R $\rightarrow$ R be defined as f(x) = det(A). Then the sum of maximum and minimum values of f on R is equal to:

(A) $-\frac{20}{27}$ (B) $\frac{88}{27}$ (C) $\frac{20}{27}$ (D) $-\frac{88}{27}$

Open solution

Q31. MCQ (Single Correct Answer)

The number of real roots of the equation $\tan^{-1}\sqrt{x(x+1)} + \sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{4}$ is :

(A) 1 (B) 2 (C) 4 (D) 0

Open solution

Q32. MCQ (Single Correct Answer)

Let y = y(x) be the solution of the differential equation $e^x\sqrt{1-y^2}dx + \left(\frac{y}{x}\right)dy = 0$, y(1) = -1. Then the value of $(y(3))^2$ is equal to :

(A) $1 - 4e^3$ (B) $1 - 4e^6$ (C) $1 + 4e^3$ (D) $1 + 4e^6$

Open solution

Q33. MCQ (Single Correct Answer)

Let 'a' be a real number such that the function f(x) = $ax^2 + 6x - 15$, x $\in$ R is increasing in $(-\infty, \frac{3}{4})$ and decreasing in $(\frac{3}{4}, \infty)$. Then the function g(x) = $ax^2 - 6x + 15$, x$\in$R has a :

(A) local maximum at x = $-\frac{3}{4}$ (B) local minimum at x = $-\frac{3}{4}$

(C) local maximum at x = $\frac{3}{4}$ (D) local minimum at x = $\frac{3}{4}$

Open solution

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Q34. MCQ (Single Correct Answer)

Let a function f : R $\to$ R be defined as $f(x) = \begin{cases} \sin x - e^x & if \quad x \le 0 \\ a + [-x] & if \quad 0 < x < 1 \\ 2x - b & if \quad x \ge 1 \end{cases}$

where [x] is the greatest integer less than or equal to x. If f is continuous on R, then (a + b) is equal to:

(A) 4 (B) 3 (C) 2 (D) 5

Open solution

Q35. MCQ (Single Correct Answer)

Words with or without meaning are to be formed using all the letters of the word EXAMINATION. The probability that the letter M appears at the fourth position in any such word is :

(A) $\frac{1}{66}$ (B) $\frac{1}{11}$ (C) $\frac{1}{9}$ (D) $\frac{2}{11}$

Open solution

Q36. MCQ (Single Correct Answer)

The probability of selecting integers a$\in$[$-$ 5, 30] such that x^2 + 2(a + 4)x $-$ 5a + 64 > 0, for all x$\in$R, is :

(A) $\frac{7}{36}$ (B) $\frac{2}{9}$ (C) $\frac{1}{6}$ (D) $\frac{1}{4}$

Open solution

Q37. Numerical

Let $\vec{a}, \vec{b}, \vec{c}$ be three mutually perpendicular vectors of the same magnitude and equally inclined at an angle θ, with the vector $\vec{a} + \vec{b} + \vec{c}$. Then 36$\cos^2 2\theta$ is equal to _____.

Open solution

Q38. Numerical

Let $A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$ and B = $7A^{20} - 20A^7 + 2I$, where I is an identity matrix of order 3 $\times$ 3. If B = $[b_{ij}]$, then b_{13} is equal to _____.

Open solution

Q39. Numerical

The number of rational terms in the binomial expansion of $\left(4^{\frac{1}{4}} + 5^{\frac{1}{6}}\right)^{120}$ is _____.

Open solution

Q40. Numerical

If the shortest distance between the lines $\vec{r_1} = \alpha\hat{i} + 2\hat{j} + 2\hat{k} + \lambda(\hat{i} - 2\hat{j} + 2\hat{k})$, $\lambda \in$ R, α > 0 and $\vec{r_2} = -4\hat{i} - \hat{k} + \mu(3\hat{i} - 2\hat{j} - 2\hat{k})$, $\mu \in$ R is 9, then α is equal to _____.

Open solution

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Q41. Numerical

Let T be the tangent to the ellipse E : $x^2 + 4y^2 = 5$ at the point P(1, 1). If the area of the region bounded by the tangent T, ellipse E, lines x = 1 and x = $\sqrt{5}$ is $\alpha\sqrt{5} + \beta + \gamma \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$, then $|\alpha + \beta + \gamma|$ is equal to _____.

Open solution

Q42. Numerical

Let a, b, c, d in arithmetic progression with common difference λ. If $\begin{vmatrix} x+a-c & x+b & x+a \\ x-1 & x+c & x+b \\ x-b+d & x+d & x+c \end{vmatrix} = 2$, then value of λ^2 is equal to _____.

Open solution

Q43. Numerical

There are 15 players in a cricket team, out of which 6 are bowlers, 7 are batsman and 2 are wicketkeepers. The number of ways, a team of 11 players be selected from them so as to include at least 4 bowlers, 5 batsman and 1 wicketkeeper, is _____.

Open solution

Q44. Numerical

If the value of $\lim\limits_{x\to 0} (2 - \cos x\sqrt{\cos 2x})^{\left(\frac{x+2}{x^2}\right)}$ is equal to e^a, then a is equal to _____.

Open solution

Physics 27 questions

Q45. MCQ (Single Correct Answer)

A deuteron and an alpha particle having equal kinetic energy enter perpendicularly into a magnetic field. Let r_d and r_α be their respective radii of circular path. The value of $\frac{r_d}{r_\alpha}$ is equal to :

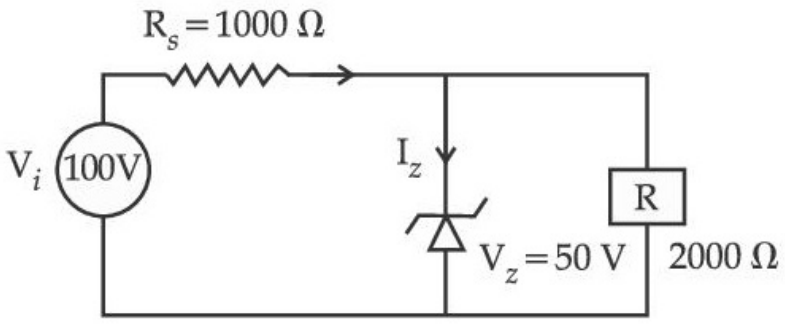
(A) 1 (B) 2 (C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}$

Open solution

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Q46. MCQ (Single Correct Answer)

For the circuit shown below, calculate the value of I_z :



Open solution

(A) 25 mA (B) 0.1 A (C) 0.15 A (D) 0.05 A

Q47. MCQ (Single Correct Answer)

A current of 5 A is passing through a non-linear magnesium wire of cross-section 0.04 m^2. At every point the direction of current density is at an angle of 60° with the unit vector of area of cross-section. The magnitude of electric field at every point of the conductor is :

(Resistivity of magnesium $\rho = 44 \times 10^{-8}\ \Omega m$)

Open solution

(A) 11×10^{-5} V/m (B) 11×10^{-3} V/m (C) 11×10^{-7} V/m (D) 11×10^{-2} V/m

Q48. MCQ (Single Correct Answer)

The normal reaction 'N' for a vehicle of 800 kg mass, negotiating a turn on a 30° banked road at maximum possible speed without skidding is _____ $\times 10^3$ kg m/s^2. [Given cos30° = 0.87, μ_s = 0.2]

Open solution

(A) 12.4 (B) 7.2 (C) 6.96 (D) 10.2

Q49. MCQ (Single Correct Answer)

Consider a mixture of gas molecule of types A, B and C having masses $m_A < m_B < m_C$. The ratio of their root mean square speeds at normal temperature and pressure is :

Open solution

(A) $v_A = v_B \neq v_C$ (B) $\frac{1}{v_A} > \frac{1}{v_B} > \frac{1}{v_C}$

(C) $\frac{1}{v_A} < \frac{1}{v_B} < \frac{1}{v_C}$ (D) $v_A = v_B = v_C = 0$

Q50. MCQ (Single Correct Answer)

The amount of heat needed to raise the temperature of 4 moles of rigid diatomic gas from 0° C to 50° C when no work is done is _____. (R is the universal gas constant).

Open solution

(A) 500 R (B) 250 R (C) 750 R (D) 175 R

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Q51. MCQ (Single Correct Answer)

A butterfly is flying with a velocity $4\sqrt{2}$ m/s in North-East direction. Wind is slowly blowing at 1 m/s from North to South. The resultant displacement of the butterfly in 3 seconds is :

Open solution

(A) $12\sqrt{2}$ m (B) 20 m (C) 3 m (D) 15 m

Q52. MCQ (Single Correct Answer)

A certain charge Q is divided into two parts q and (Q − q). How should the charges Q and q be divided so that q and (Q − q) placed at a certain distance apart experience maximum electrostatic repulsion?

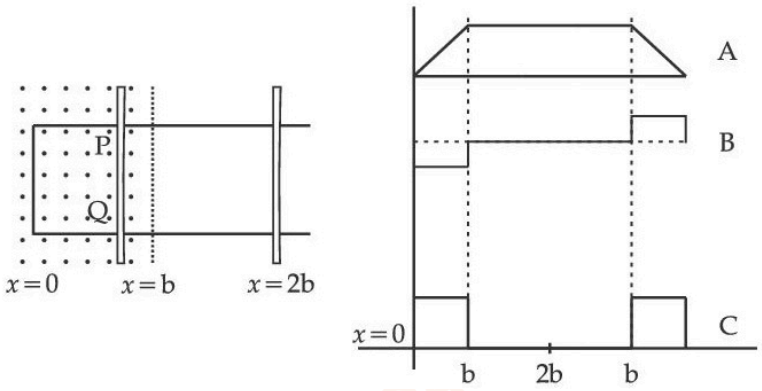
Open solution

(A) Q = 2q (B) Q = 4q (C) Q = 3q (D) Q = $\frac{q}{2}$

Q53. MCQ (Single Correct Answer)

The arm PQ of a rectangular conductor is moving from x = 0 to x = 2b outwards and then inwards from x = 2b to x = 0 as shown in the figure. A uniform magnetic field perpendicular to the plane is acting from x = 0 to x = b. Identify the graph showing the variation of different quantities with distance.

Open solution

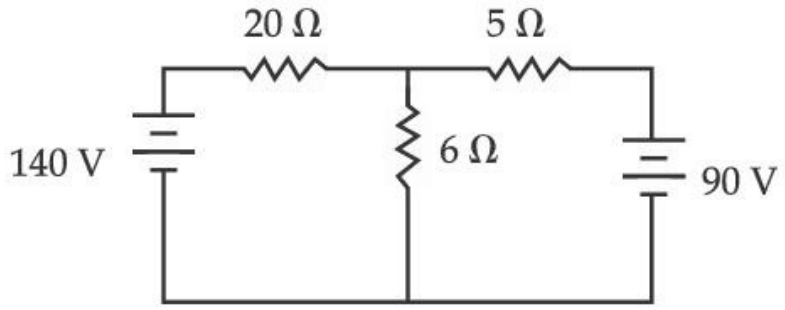


(A) A-Flux, B-Power dissipated, C-EMF (B) A-Flux, B-EMF, C-Power dissipated

(C) A-Power dissipated, B-Flux, C-EMF (D) A-EMF, B-Power dissipated, C-Flux

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Q54. MCQ (Single Correct Answer)



Open solution

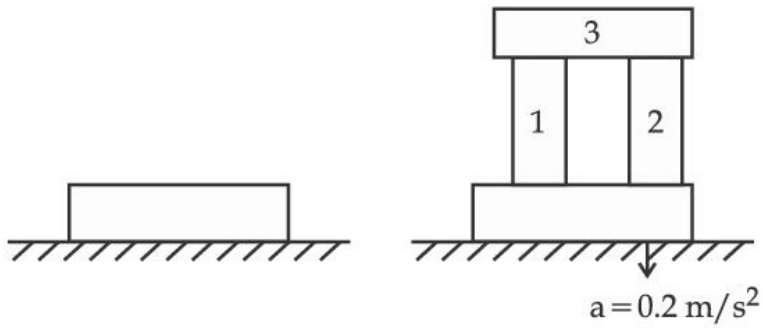
The value of current in the 6 Ω resistance is :

(A) 4A (B) 8A (C) 10A (D) 6A

Q55. MCQ (Single Correct Answer)

A steel block of 10 kg rests on a horizontal floor as shown. When three iron cylinders are placed on it as shown, the block and cylinders go down with an acceleration 0.2 m/s^2. The normal reaction R' by the floor if mass of the iron cylinders are equal and of 20 kg each, is _____ N. [Take g = 10 m/s^2 and μ_s = 0.2]

Open solution



(A) 686 (B) 684 (C) 714 (D) 716

Q56. MCQ (Single Correct Answer)

A nucleus of mass M emits γ -ray photon of frequency 'v'. The loss of internal energy by the nucleus is :

Open solution

[Take 'c' as the speed of electromagnetic wave]

(A) hv (B) $hv\left[1+\frac{hv}{2Mc^2}\right]$

(C) $hv\left[1-\frac{hv}{2Mc^2}\right]$ (D) 0

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Q57. MCQ (Single Correct Answer)

The radiation corresponding to $3 \rightarrow 2$ transition of a hydrogen atom falls on a gold surface to generate photoelectrons. The electrons are passed through a magnetic field of 5×10^{-4} T. Assume that the radius of the largest circular path followed by these electrons is 7 mm, the work function of the metal is : (Mass of electron = 9.1×10^{-31} kg)

Open solution

(A) 1.36 eV (B) 1.88 eV (C) 0.82 eV (D) 0.16 eV

Q58. MCQ (Single Correct Answer)

If $\vec{A}$ and $\vec{B}$ are two vectors satisfying the relation $\vec{A} \cdot \vec{B} = \left|\vec{A} \times \vec{B}\right|$. Then the value of $\left|\vec{A} - \vec{B}\right|$ will be :

Open solution

(A) $\sqrt{A^2 + B^2 + \sqrt{2}AB}$ (B) $\sqrt{A^2 + B^2}$

(C) $\sqrt{A^2 + B^2 - \sqrt{2}AB}$ (D) $\sqrt{A^2 + B^2 + 2AB}$

Q59. MCQ (Single Correct Answer)

The entropy of any system is given by
$S = \alpha^2\beta \ln\left[\frac{\mu kR}{J\beta^2} + 3\right]$ where α and β are the constants. μ, J, k and R are no. of moles, mechanical equivalent of heat, Boltzmann constant and gas constant respectively.
[Take $S = \frac{dQ}{T}$]
Choose the incorrect option from the following :

Open solution

(A) α and J have the same dimensions.

(B) S and α have different dimensions

(C) S, β, k and μR have the same dimensions

(D) α and k have the same dimensions

Q60. MCQ (Single Correct Answer)

AC voltage V(t) = 20 sinωt of frequency 50 Hz is applied to a parallel plate capacitor. The separation between the plates is 2 mm and the area is 1 m^2. The amplitude of the oscillating displacement current for the applied AC voltage is _____. [Take $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m]

Open solution

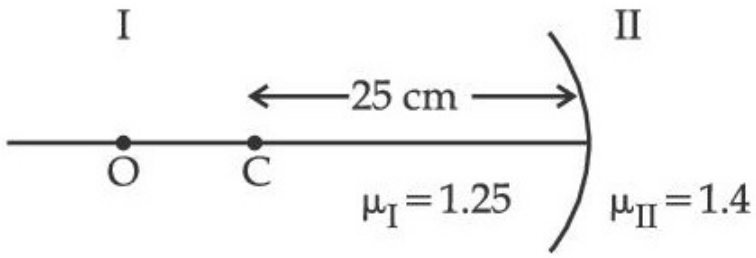
(A) 55.58 μA (B) 21.14 μA (C) 27.79 μA (D) 83.37 μA

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Q61. MCQ (Single Correct Answer)

Region I and II are separated by a spherical surface of radius 25 cm. An object is kept in region I at a distance of 40 cm from the surface. The distance of the image from the surface is :

Open solution

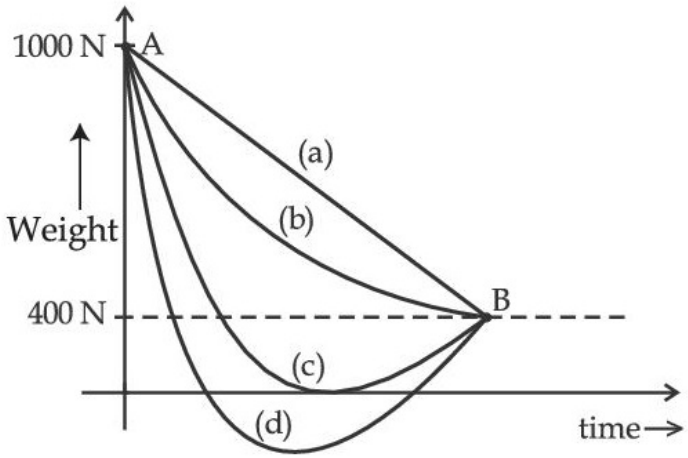


(A) 55.44 cm (B) 18.23 cm (C) 9.52 cm (D) 37.58 cm

Q62. MCQ (Single Correct Answer)

A person whose mass is 100 kg travels from Earth to Mars in a spaceship. Neglect all other objects in sky and take acceleration due to gravity on the surface of the Earth and Mars as 10 m/s^2 and 4 m/s^2 respectively. Identify from the below figures, the curve that fits best for the weight of the passenger as a function of time.

Open solution



(A) (b) (B) (c) (C) (d) (D) (a)

Q63. MCQ (Single Correct Answer)

The value of tension in a long thin metal wire has been changed from T_1 to T_2. The lengths of the metal wire at two different values of tension T_1 and T_2 are l_1 and l_2 respectively. The actual length of the metal wire is :

Open solution

(A) $\frac{l_1+l_2}{2}$ (B) $\sqrt{T_1T_2l_1l_2}$

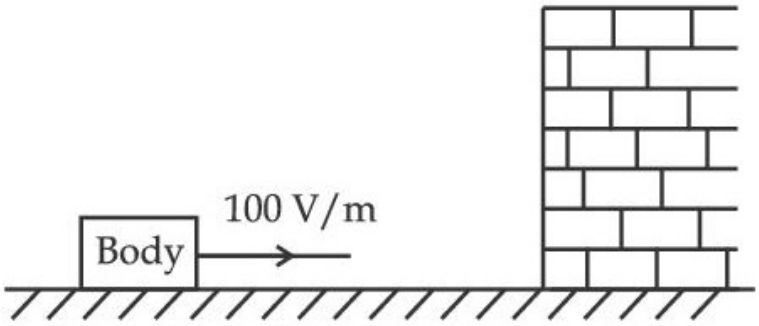
(C) $\frac{T_1l_2-T_2l_1}{T_1-T_2}$ (D) $\frac{T_1l_1-T_2l_2}{T_1-T_2}$

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Q64. Numerical

A body having specific charge 8 μC/g is resting on a frictionless plane at a distance 10 cm from the wall (as shown in the figure). It starts moving towards the wall when a uniform electric field of 100 V/m is applied horizontally towards the wall. If the collision of the body with the wall is perfectly elastic, then the time period of the motion will be _____ s.

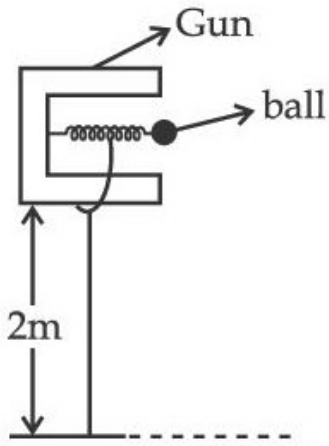
Open solution



Q65. Numerical

In a spring gun having spring constant 100 N/m a small ball 'B' of mass 100 g is put in its barrel (as shown in figure) by compressing the spring through 0.05 m. There should be a box placed at a distance 'd' on the ground so that the ball falls in it. If the ball leaves the gun horizontally at a height of 2 m above the ground. The value of d is _____ m. (g = 10 m/s^2).

Open solution



Q66. Numerical

An object viewed from a near point distance of 25 cm, using a microscopic lens with magnification '6', gives an unresolved image. A resolved image is observed at infinite distance with a total magnification double the earlier using an eyepiece along with the given lens and a tube of length 0.6 m, if the focal length of the eyepiece is equal to _____ cm.

Open solution

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Q67. Numerical

A circular disc reaches from top to bottom of an inclined plane of length 'L'. When it slips down the plane, it makes time 't_1'. When it rolls down the plane, it takes time t_2. The value of $\frac{t_2}{t_1}$ is $\sqrt{\frac{3}{x}}$. The value of x will be _____.

Open solution

Q68. Numerical

The amplitude of wave disturbance propagating in the positive x-direction is given by $y = \frac{1}{1+x^2}$ at time t = 0 and $y = \frac{1}{1+(x-2)^2}$ at t = 1 s, where x and y are in metres. The shape of wave does not change during the propagation. The velocity of the wave will be _____ m/s.

Open solution

Q69. Numerical

In an LCR series circuit, an inductor 30 mH and a resistor 1 Ω are connected to an AC source of angular frequency 300 rad/s. The value of capacitance for which, the current leads the voltage by 45° is $\frac{1}{x} \times 10^{-3}$ F. Then the value of x is _____.

Open solution

Q70. Numerical

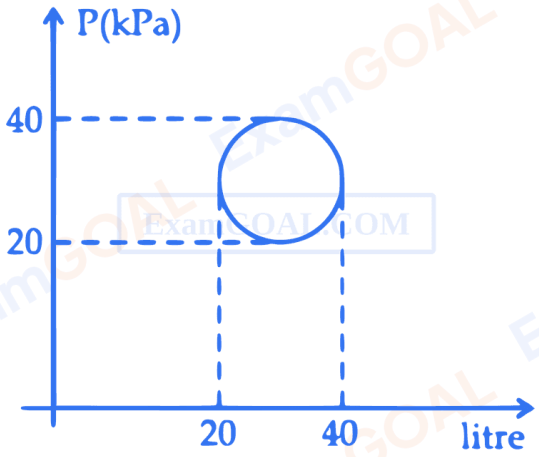
A rod of mass M and length L is lying on a horizontal frictionless surface. A particle of mass 'm' travelling along the surface hits at one end of the rod with a velocity 'u' in a direction perpendicular to the rod. The collision is completely elastic. After collision, particle comes to rest. The ratio of masses $\left(\frac{m}{M}\right)$ is $\frac{1}{x}$. The value of 'x' will be _____.

Open solution

Q71. Numerical

In the reported figure, heat energy absorbed by a system in going through a cyclic process is _____ πJ.

Open solution



Out of syllabus 19 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q72. MCQ (Single Correct Answer) · **Chemistry**

Green chemistry in day-to-day life is in the use of :

(A) Chlorine for bleaching of paper

(B) Large amount of water alone for washing clothes

(C) Tetrachloroethene for laundry

(D) Liquified CO_2 for dry cleaning of clothes

Open solution

Q73. MCQ (Single Correct Answer) · **Chemistry**

The set in which compounds have different nature is :

(A) $B(OH)_3$ and H_3PO_3

(B) $B(OH)_3$ and $Al(OH)_3$

(C) NaOH and $Ca(OH)_2$

(D) $Be(OH)_2$ and $Al(OH)_3$

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Sharp glass edge becomes smooth on heating it upto its melting point.

Reason R : The viscosity of glass decreases on melting.

Choose the most appropriate answer from the options given below.

(A) A is true but R is false

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is false but R is true.

(D) Both A and R are true and R is the correct explanation of A.

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

Orlon fibres are made up of :

(A) Polyacrylonitrile

(B) Polyesters

(C) Polyamide

(D) Cellulose

Open solution

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Q76. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements : One is labelled as Assertion A and other is labelled as Reason R.

Assertion A : The dihedral angles in H_2O_2 in gaseous phase is 90.2° and in solid phase is 111.5°.

Reason R : The change in dihedral angle in solid and gaseous phase is due to the difference in the intermolecular forces.

Choose the most appropriate answer from the options given below for A and R.

(A) A is correct but R is not correct.

(B) Both A and R are correct but R is not the correct explanation of A.

(C) Both A and R are correct and R is the correct explanation of A.

(D) A is not correct but R is correct.

Q77. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Chemical nature of the nitrogen oxide compound obtained from a reaction of concentrated nitric acid and P_4O_{10} (in 4 : 1 ratio) is :

(A) acidic **(B)** basic **(C)** amphoteric **(D)** neutral

Q78. MCQ (Single Correct Answer) · **Chemistry**

Open solution

A s-block element (M) reacts with oxygen to form an oxide of the formula MO_2. The oxide is pale yellow in colour and paramagnetic. The element (M) is :

(A) Mg **(B)** Na **(C)** Ca **(D)** K

Q79. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The metal that can be purified economically by fractional distillation method is :

(A) Fe **(B)** Zn **(C)** Cu **(D)** Ni

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Q80. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The conditions given below are in the context of observing Tyndall effect in colloidal solutions :

(A) The diameter of the colloidal particles is comparable to the wavelength of light used.

(B) The diameter of the colloidal particles is much smaller than the wavelength of light used.

(C) The diameter of the colloidal particles is much larger than the wavelength of light used.

(D) The refractive indices of the dispersed phase and the dispersion medium are comparable.

(E) The dispersed phase has a very different refractive index from the dispersion medium.

Choose the most appropriate conditions from the options given below :

(A) (A) and (E) only **(B)** (C) and (D) only **(C)** (A) and (D) only **(D)** (B) and (E) only

Q81. MCQ (Single Correct Answer) · **Mathematics**

Open solution

The Boolean expression $(p \wedge \sim q) \Rightarrow (q \vee \sim p)$ is equivalent to :

(A) $q \Rightarrow p$ **(B)** $p \Rightarrow q$ **(C)** $\sim q \Rightarrow p$ **(D)** $p \Rightarrow \sim q$

Q82. MCQ (Single Correct Answer) · **Mathematics**

Open solution

If in a triangle ABC, AB = 5 units, $\angle B = \cos^{-1}\left(\frac{3}{5}\right)$ and radius of circumcircle of ΔABC is 5 units, then the area (in sq. units) of ΔABC is :

(A) $10 + 6\sqrt{2}$ **(B)** $8 + 2\sqrt{2}$ **(C)** $6 + 8\sqrt{3}$ **(D)** $4 + 2\sqrt{3}$

Q83. MCQ (Single Correct Answer) · **Mathematics**

Open solution

The coefficient of x^{256} in the expansion of

$(1 - x)^{101} (x^2 + x + 1)^{100}$ is :

(A) ${}^{100}C_{16}$ **(B)** ${}^{100}C_{15}$ **(C)** $-{}^{100}C_{16}$ **(D)** $-{}^{100}C_{15}$

Q84. MCQ (Single Correct Answer) · **Mathematics**

Open solution

Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $\vec{a}.\vec{c} = |\vec{c}|, |\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $(\vec{a} \times \vec{b})$ and $\vec{c}$ is $\frac{\pi}{6}$, then the value of $\left|\left(\vec{a} \times \vec{b}\right) \times \vec{c}\right|$ is :

(A) $\frac{2}{3}$ **(B)** 4 **(C)** 3 **(D)** $\frac{3}{2}$

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Q85. MCQ (Single Correct Answer) · **Mathematics**

Let the tangent to the parabola S : $y^2 = 2x$ at the point P(2, 2) meet the x-axis at Q and normal at it meet the parabola S at the point R. Then the area (in sq. units) of the triangle PQR is equal to :

(A) $\frac{25}{2}$ (B) $\frac{35}{2}$ (C) $\frac{15}{2}$ (D) 25

Open solution

Q86. Numerical · **Mathematics**

Let P be a plane passing through the points (1, 0, 1), (1, −2, 1) and (0, 1, −2). Let a vector $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ be such that $\vec{a}$ is parallel to the plane P, perpendicular to $(\hat{i} + 2\hat{j} + 3\hat{k})$ and $\vec{a}.\ (\hat{i} + \hat{j} + 2\hat{k}) = 2$, then $(\alpha - \beta + \gamma)^2$ equals _____.

Open solution

Q87. Numerical · **Mathematics**

Let y = mx + c, m > 0 be the focal chord of $y^2 = -64x$, which is tangent to $(x + 10)^2 + y^2 = 4$. Then, the value of $4\sqrt{2}$ (m + c) is equal to _____.

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

A radioactive material decays by simultaneous emissions of two particles with half lives of 1400 years and 700 years respectively. What will be the time after which one third of the material remains ? (Take ln 3 = 1.1)

(A) 740 years (B) 1110 years (C) 700 years (D) 340 years

Open solution

Q89. Numerical · **Physics**

A carrier wave $V_c(t) = 160 \sin(2\pi \times 10^6 t)$ volts is made to vary between V_{max} = 200 V and V_{min} = 120 V by a message signal $V_m(t) = A_m \sin(2\pi \times 10^3 t)$ volts. The peak voltage A_m of the modulating signal is _____.

Open solution

Q90. Numerical · **Physics**

The frequency of a car horn encountered a change from 400 Hz to 500 Hz, when the car approaches a vertical wall. If the speed of sound is 330 m/s. Then the speed of car is _____ km/h.

Open solution

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Answer Key

Chemistry

1. **B**	2. **D**	3. **C**	4. **A**	5. **C**
6. **C**	7. **C**	8. **C**	9. **A**	10. **C**
11. **D**	12. **3**	13. **226**	14. **0**	15. **2**
16. **172**	17. **3**	18. **2**	19. **106**	20. **667**
21. **78**				

Mathematics

22. **B**	23. **A**	24. **C**	25. **C**	26. **A**
27. **B**	28. **C**	29. **A**	30. **D**	31. **D**
32. **B**	33. **A**	34. **B**	35. **B**	36. **B**
37. **4**	38. **910**	39. **21**	40. **6**	41. **1.25**
42. **1**	43. **777**	44. **3**		

Physics

45. **C**	46. **A**	47. **A**	48. **D**	49. **C**
50. **A**	51. **D**	52. **A**	53. **B**	54. **C**
55. **A**	56. **B**	57. **C**	58. **C**	59. **D**
60. **C**	61. **D**	62. **B**	63. **C**	64. **1**
65. **1**	66. **25**	67. **2**	68. **2**	69. **3**
70. **4**	71. **100**			

Out of syllabus

72. **D**	73. **B**	74. **B**	75. **A**	76. **D**
77. **A**	78. **D**	79. **B**	80. **A**	81. **B**
82. **C**	83. **B**	84. **D**	85. **A**	86. **81**
87. **34**	88. **A**	89. **40**	90. **132**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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